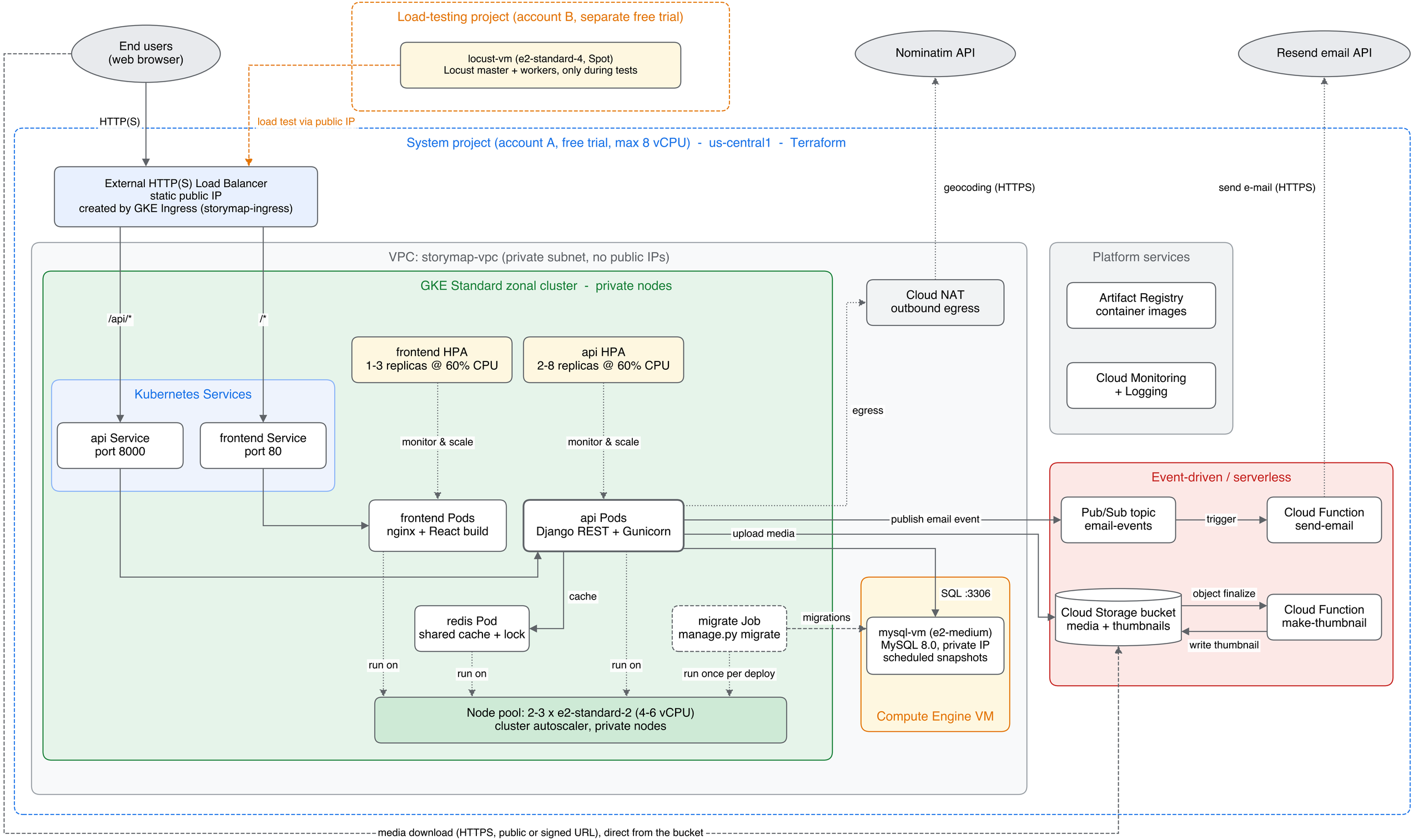
Load-testing project (account B, separate free trial)
locust-vm (e2-standard-4, Spot)
Locust master + workers, only during tests
End users
(web browser)
HTTP(S)
load test via public IP
Nominatim API
Resend email API
System project (account A, free trial, max 8 vCPU) - us-central1 - Terraform
External HTTP(S) Load Balancer
static public IP
created by GKE Ingress (storymap-ingress)
geocoding (HTTPS)
send e-mail (HTTPS)
VPC: storymap-vpc (private subnet, no public IPs)
GKE Standard zonal cluster - private nodes
/api/*
/*
Kubernetes Services
api Service
port 8000
frontend Service
port 80
frontend HPA
1-3 replicas @ 60% CPU
api HPA
2-8 replicas @ 60% CPU
monitor & scale
monitor & scale
frontend Pods
nginx + React build
api Pods
Django REST + Gunicorn
cache
redis Pod
shared cache + lock
migrate Job
manage.py migrate
migrations
run on
run on
run on
run once per deploy
Node pool: 2-3 x e2-standard-2 (4-6 vCPU)
cluster autoscaler, private nodes
Cloud NAT
outbound egress
egress
SQL :3306
mysql-vm (e2-medium)
MySQL 8.0, private IP
scheduled snapshots
Compute Engine VM
Platform services
Artifact Registry
container images
Cloud Monitoring
+ Logging
publish email event
upload media
Event-driven / serverless
Pub/Sub topic
email-events
trigger
Cloud Function
send-email
Cloud Storage bucket
media + thumbnails
object finalize
write thumbnail
Cloud Function
make-thumbnail
media download (HTTPS, public or signed URL), direct from the bucket